Markov chain Monte Carlo

Lecture 5

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The idea of MCMC

We want to learn about unknown quantities $\boldsymbol{w}$ from their posterior:

$$\pi(\boldsymbol{w}) = p(\boldsymbol{w} \mid x) \propto p(x, \boldsymbol{w}).$$

Directly drawing from this distribution can be difficult.

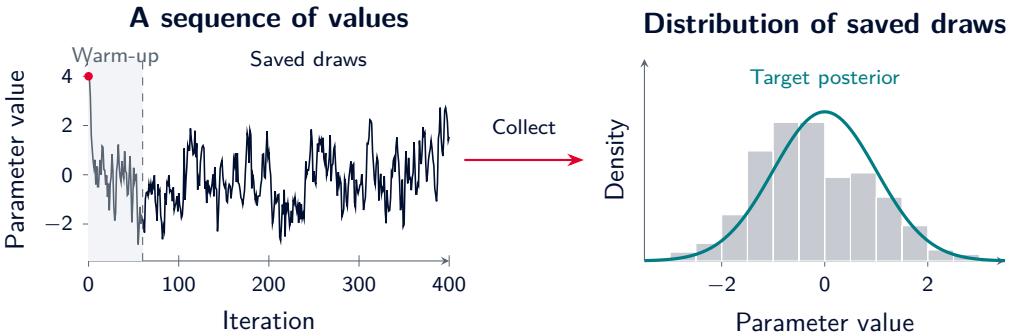
- **VB** fits a simpler distribution to approximate the posterior.
- **MCMC** generates a sequence of values that explores the posterior.
- Each new value uses the current value. This creates a **Markov chain**.
- Use the saved draws to estimate posterior means, intervals, and predictions.

MCMC targets the posterior itself. A finite run still has simulation error.

Reading: Gelman et al., *Bayesian Data Analysis*, 3rd ed., Chapter 11.

How MCMC works

Start with an initial value, then repeatedly apply a random update.



Discard warm-up draws, then use the saved values for posterior means and intervals.

Why MCMC works

The update rule has two essential features:

- **It preserves the posterior.** If the current state already followed the posterior distribution, one more update would leave that distribution unchanged.
- **It allows exploration.** The chain must be able to move throughout the posterior, and we must run it long enough to explore the relevant regions.

For a well-explored run, a region with 30% posterior probability should contain about 30% of the saved draws.

How often the chain visits a region tells us its posterior probability.

Gibbs sampling

Gibbs is useful when we can draw from each block's **conditional posterior**. Partition the unknowns into d blocks:

$$\boldsymbol{w} = (w_1, \ldots, w_d).$$

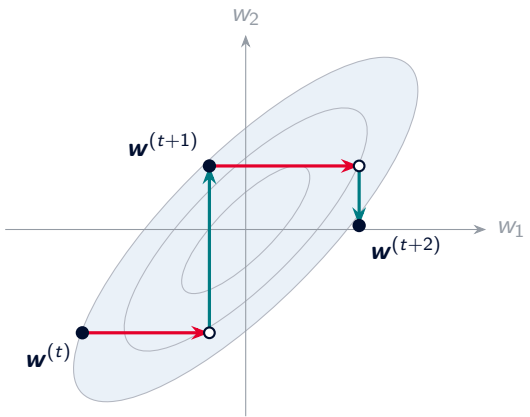
At iteration t, one full update (a **sweep**) visits $j = 1, \ldots, d$ in order and draws

$$w_j^{(t+1)} \sim \pi\Big(\cdot \,\Big|\, w_1^{(t+1)}, \ldots, w_{j-1}^{(t+1)}, w_{j+1}^{(t)}, \ldots, w_d^{(t)}\Big).$$

- For blocks already updated, use their **new values**.
- For blocks not yet updated, use their values from iteration t.

Record $\boldsymbol{w}^{(t+1)} = (w_1^{(t+1)}, \ldots, w_d^{(t+1)})$ and repeat.

Gibbs updates in 2D



Ellipses: contours of the joint posterior

1. Horizontal move

Hold w_2 fixed and sample w_1:

$$w_1^{(t+1)} \sim \pi(\cdot \mid w_2^{(t)}).$$

2. Vertical move

Use the new w_1 and sample w_2:

$$w_2^{(t+1)} \sim \pi(\cdot \mid w_1^{(t+1)}).$$

The two moves complete one **sweep**.
Repeat from the new state.

Solid dots: complete states.
Open dots: intermediate states.

Metropolis–Hastings

Choose a starting state $\boldsymbol{w}^{(0)}$ with positive posterior density. At iteration t:

1. **Propose:** draw $\boldsymbol{w}^{\star} \sim q(\cdot \mid \boldsymbol{w}^{(t)})$.
2. **Compute:** the probability of accepting this proposal is

$$\boxed{\alpha = \min\left\{1, \frac{\pi(\boldsymbol{w}^{\star}) q(\boldsymbol{w}^{(t)} \mid \boldsymbol{w}^{\star})}{\pi(\boldsymbol{w}^{(t)}) q(\boldsymbol{w}^{\star} \mid \boldsymbol{w}^{(t)})}\right\}.}$$

3. **Accept or stay:** draw $a \sim \text{Uniform}(0, 1)$ independently.

$$\boldsymbol{w}^{(t+1)} = \begin{cases} \boldsymbol{w}^{\star}, & a \leq \alpha, \\ \boldsymbol{w}^{(t)}, & a > \alpha. \end{cases}$$

Record one state every iteration, including repeated states after rejection.
We may replace $\pi(\boldsymbol{w})$ by $p(x, \boldsymbol{w})$ because the normalizing constant cancels.

A random-walk proposal

A common choice adds Gaussian noise to the current state:

$$\boldsymbol{w}^{\star} = \boldsymbol{w}^{(t)} + b\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \quad b > 0.$$

This proposal is symmetric, so the acceptance rule simplifies to

$$\alpha = \min\left\{1, \frac{\pi(\boldsymbol{w}^{\star})}{\pi(\boldsymbol{w}^{(t)})}\right\}.$$

Proposed density / current density	Acceptance probability
2	1 (always accept)
0.4	0.4 (accept with probability 40%)

Step size b**:** too small gives tiny moves, too large gives many rejections.
Tune b during warm-up, then fix it when collecting draws.

MH illustration



Solid dots: recorded states
Open dots: rejected proposals

Target: $\pi(w) = \mathcal{N}(w; 0, 1)$.

Use a symmetric proposal.

Draw $a \sim \text{Uniform}(0, 1)$.
Accept if $a \leq \alpha$.

Iteration	α	a	Decision
1	1	0.35	Accept
2	0.22	0.60	Reject
3	0.82	0.40	Accept

Rejections keep the current value:

$$\boxed{w^{(2)} = w^{(1)} = -0.5.}$$

Burn-in and thinning

Burn-in (warm-up): discard the first B iterations to reduce the influence of the starting values.

$$\underbrace{\boldsymbol{w}^{(1)}, \ldots, \boldsymbol{w}^{(B)}}_{\text{discard}} \qquad \underbrace{\boldsymbol{w}^{(B+1)}, \ldots, \boldsymbol{w}^{(T)}}_{\text{use for inference}}.$$

Thinning: keep only every Lth draw after warm-up.

Example: 10,000 iterations, 2,000 warm-up	Draws kept
Keep every remaining draw ($L = 1$)	8,000
Keep every fifth draw ($L = 5$)	1,600

Usually keep all draws after warm-up. Thinning can reduce storage needs.
Neither discarding early draws nor thinning makes the saved draws independent.

Using the saved draws

Suppose we save S draws after warm-up: $\widetilde{\boldsymbol{w}}^{(1)}, \ldots, \widetilde{\boldsymbol{w}}^{(S)}$.

- **Posterior mean:** average the quantity of interest over the draws.

$$\mathbb{E}[h(\boldsymbol{w}) \mid x] \approx \frac{1}{S} \sum_{s=1}^{S} h(\widetilde{\boldsymbol{w}}^{(s)}).$$

- **95% credible interval:** use the 2.5% and 97.5% sample quantiles.

Run several chains from different starting values and inspect plots of the draws against iteration number (**trace plots**).

Example 3 revisited

Each observation belongs to one of K Gaussian components:

$$\mu_k \sim \mathcal{N}(0, \tau^2), \qquad k = 1, \ldots, K,$$

$$c_i \sim \text{Categorical}(1/K, \ldots, 1/K), \qquad i = 1, \ldots, n,$$

$$X_i \mid c_i = k, \mu \sim \mathcal{N}(\mu_k, 1).$$

- c_i is the component label for observation i. μ_k is component k's mean.
- We know K, τ^2, the equal weights, and the observation variance 1.

Gibbs algorithm alternates between labels c and means μ. The full state is $w = (c, \mu)$.

Step 1: sample labels

For one observation, Bayes' rule gives

$$\begin{aligned}\Pr(c_i = k \mid \mu, x) &\propto \Pr(c_i = k)\, p(x_i \mid c_i = k, \mu) \\ &= \frac{1}{K}\frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(x_i - \mu_k)^2\right\}.\end{aligned}$$

The common constants cancel when we normalize over k:

$$\boxed{\Pr(c_i = k \mid \mu, x) = \omega_{ik}(\mu) = \frac{\exp\{-(x_i - \mu_k)^2/2\}}{\sum_{\ell=1}^{K} \exp\{-(x_i - \mu_\ell)^2/2\}}.}$$

Therefore, we draw a label for each observation:

$$c_i \mid \mu, x \sim \text{Categorical}(\omega_{i1}, \ldots, \omega_{iK}).$$

Step 2: sample means

For a fixed assignment vector c, let $n_k = \sum_i \mathbf{1}\{c_i = k\}$ and $T_k = \sum_i \mathbf{1}\{c_i = k\} x_i$.

$$
\begin{aligned}
p(\mu_k \mid c, x) &\propto \exp\left\{ -\frac{\mu_k^2}{2\tau^2} - \frac{1}{2} \sum_{i : c_i = k} (x_i - \mu_k)^2 \right\} \\
&\propto \exp\left[-\frac{1}{2} \left\{ (\tau^{-2} + n_k)\mu_k^2 - 2 T_k \mu_k \right\} \right] \\
&\propto \exp\left\{ -\frac{(\mu_k - a_k)^2}{2 v_k} \right\},
\end{aligned}
$$

where

$$
\boxed{a_k = \frac{T_k}{\tau^{-2} + n_k}, \qquad v_k = \frac{1}{\tau^{-2} + n_k}.}
$$

Thus $p(\mu \mid c, x) = \prod_{k=1}^{K} \mathcal{N}(\mu_k; a_k, v_k)$.

Gibbs for Example 3

Input: $x_{1:n}$, K, τ^2, total sweeps T, and warm-up length B.

Initialize: choose starting means $\mu^{(0)}$.

For $t = 0, \dots, T-1$:

1. **Draw labels** using the current means $\mu^{(t)}$:

$$c_i^{(t+1)} \sim \text{Categorical}\{\omega_{i1}(\mu^{(t)}), \dots, \omega_{iK}(\mu^{(t)})\}, \quad i = 1, \dots, n.$$

2. **Recalculate** n_k and T_k using the **new labels** $c^{(t+1)}$.

3. **Draw means** from the resulting normal distributions:

$$\mu_k^{(t+1)} \sim \mathcal{N}\Big\{ a_k(c^{(t+1)}), v_k(c^{(t+1)}) \Big\}, \quad k = 1, \dots, K.$$

Record $(c^{(t+1)}, \mu^{(t+1)})$ after each sweep.

Use sweeps $B+1, \dots, T$ for posterior summaries.

Appendix

Markov chains

Define the **transition law** by

$$\boldsymbol{W}^{(t+1)} \mid \boldsymbol{W}^{(t)} = \boldsymbol{w}^{(t)} \sim P(\cdot \mid \boldsymbol{w}^{(t)}).$$

The **Markov property** states that

$$P(\boldsymbol{w}^{(t+1)} \mid \boldsymbol{w}^{(0)}, \ldots, \boldsymbol{w}^{(t)}) = P(\boldsymbol{w}^{(t+1)} \mid \boldsymbol{w}^{(t)}).$$

For a time-homogeneous chain, abbreviate this as $P(\boldsymbol{w}' \mid \boldsymbol{w})$, with $\boldsymbol{w}$ the current state and $\boldsymbol{w}'$ the next state.

A target density π is **stationary** if

$$\boxed{\int \pi(\boldsymbol{w}) P(\boldsymbol{w}' \mid \boldsymbol{w}) \, d\boldsymbol{w} = \pi(\boldsymbol{w}').}$$

Equivalently, $\boldsymbol{W}^{(t)} \sim \pi$ implies $\boldsymbol{W}^{(t+1)} \sim \pi$.

Gibbs: stationarity

For $\boldsymbol{w} = (w_1, w_2)$ and $\boldsymbol{w}' = (w_1', w_2')$, the **full-sweep transition** is

$$P(\boldsymbol{w}' \mid \boldsymbol{w}) = \pi(w_1' \mid w_2)\pi(w_2' \mid w_1').$$

If the current state has density π, the next-state density is

$$\begin{aligned} g_{t+1}(\boldsymbol{w}') &= \iint \pi(w_1, w_2)\pi(w_1' \mid w_2)\pi(w_2' \mid w_1')\, dw_1\, dw_2 \\ &= \pi(w_2' \mid w_1') \int \pi(w_1' \mid w_2) \underbrace{\left[\int \pi(w_1, w_2)\, dw_1\right]}_{\pi_2(w_2)} dw_2 \\ &= \pi(w_2' \mid w_1') \int \pi(w_1', w_2)\, dw_2 \\ &= \pi(w_2' \mid w_1')\pi_1(w_1') = \boxed{\pi(\boldsymbol{w}')}. \end{aligned}$$

MH: transition law

Define the **accepted-move density** from $\boldsymbol{w}$ to $\boldsymbol{w}'$:

$$\boxed{k(\boldsymbol{w}' \mid \boldsymbol{w}) = q(\boldsymbol{w}' \mid \boldsymbol{w})\alpha(\boldsymbol{w}, \boldsymbol{w}').}$$

Its integral is the acceptance probability, so the **rejection probability** is

$$\rho(\boldsymbol{w}) = 1 - \int k(\boldsymbol{w}' \mid \boldsymbol{w})\, d\boldsymbol{w}'.$$

The **full transition law** is

$$\boxed{P(\boldsymbol{w}' \mid \boldsymbol{w}) = \underbrace{k(\boldsymbol{w}' \mid \boldsymbol{w})}_{\text{accepted moves}} + \underbrace{\rho(\boldsymbol{w})\delta(\boldsymbol{w}' = \boldsymbol{w})}_{\text{rejections}}.}$$

Here $\delta(\boldsymbol{w}' = \boldsymbol{w})$ is a unit point mass at $\boldsymbol{w}' = \boldsymbol{w}$.

MH: detailed balance

For accepted moves, the acceptance rule gives

$$\begin{aligned}\pi(\boldsymbol{w})k(\boldsymbol{w}' \mid \boldsymbol{w}) &= \pi(\boldsymbol{w})q(\boldsymbol{w}' \mid \boldsymbol{w}) \min\left\{1, \frac{\pi(\boldsymbol{w}')q(\boldsymbol{w} \mid \boldsymbol{w}')}{\pi(\boldsymbol{w})q(\boldsymbol{w}' \mid \boldsymbol{w})}\right\} \\ &= \min\{\pi(\boldsymbol{w})q(\boldsymbol{w}' \mid \boldsymbol{w}),\, \pi(\boldsymbol{w}')q(\boldsymbol{w} \mid \boldsymbol{w}')\} \\ &= \pi(\boldsymbol{w}')k(\boldsymbol{w} \mid \boldsymbol{w}').\end{aligned}$$

The symmetric minimum shows that accepted moves balance in both directions.

MH: stationarity

Suppose the current state has density π. Let g_{t+1} denote the **marginal density of the next state**. Then

$$
\begin{aligned}
g_{t+1}(\boldsymbol{w}') &= \int \pi(\boldsymbol{w}) P(\boldsymbol{w}' \mid \boldsymbol{w})\, d\boldsymbol{w} \\
&= \int \pi(\boldsymbol{w}) \left\{ k(\boldsymbol{w}' \mid \boldsymbol{w}) + \rho(\boldsymbol{w}) \delta(\boldsymbol{w}' = \boldsymbol{w}) \right\} d\boldsymbol{w} \\
&= \int \pi(\boldsymbol{w}) k(\boldsymbol{w}' \mid \boldsymbol{w})\, d\boldsymbol{w} + \pi(\boldsymbol{w}') \rho(\boldsymbol{w}') \\
&= \pi(\boldsymbol{w}') \int k(\boldsymbol{w} \mid \boldsymbol{w}')\, d\boldsymbol{w} + \pi(\boldsymbol{w}') \rho(\boldsymbol{w}') \\
&= \pi(\boldsymbol{w}') \{ 1 - \rho(\boldsymbol{w}') + \rho(\boldsymbol{w}') \} \\
&= \boxed{\pi(\boldsymbol{w}')}.
\end{aligned}
$$

Reading

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*, 3rd edition. Chapter 11, *Basics of Markov chain simulation*.

Section 11.1	Gibbs sampling, pp. 276–278
Sections 11.2–11.3	Metropolis–Hastings and block updates, pp. 278–281
Section 11.4 (selected parts)	Posterior summaries, warm-up, and thinning, pp. 281–283